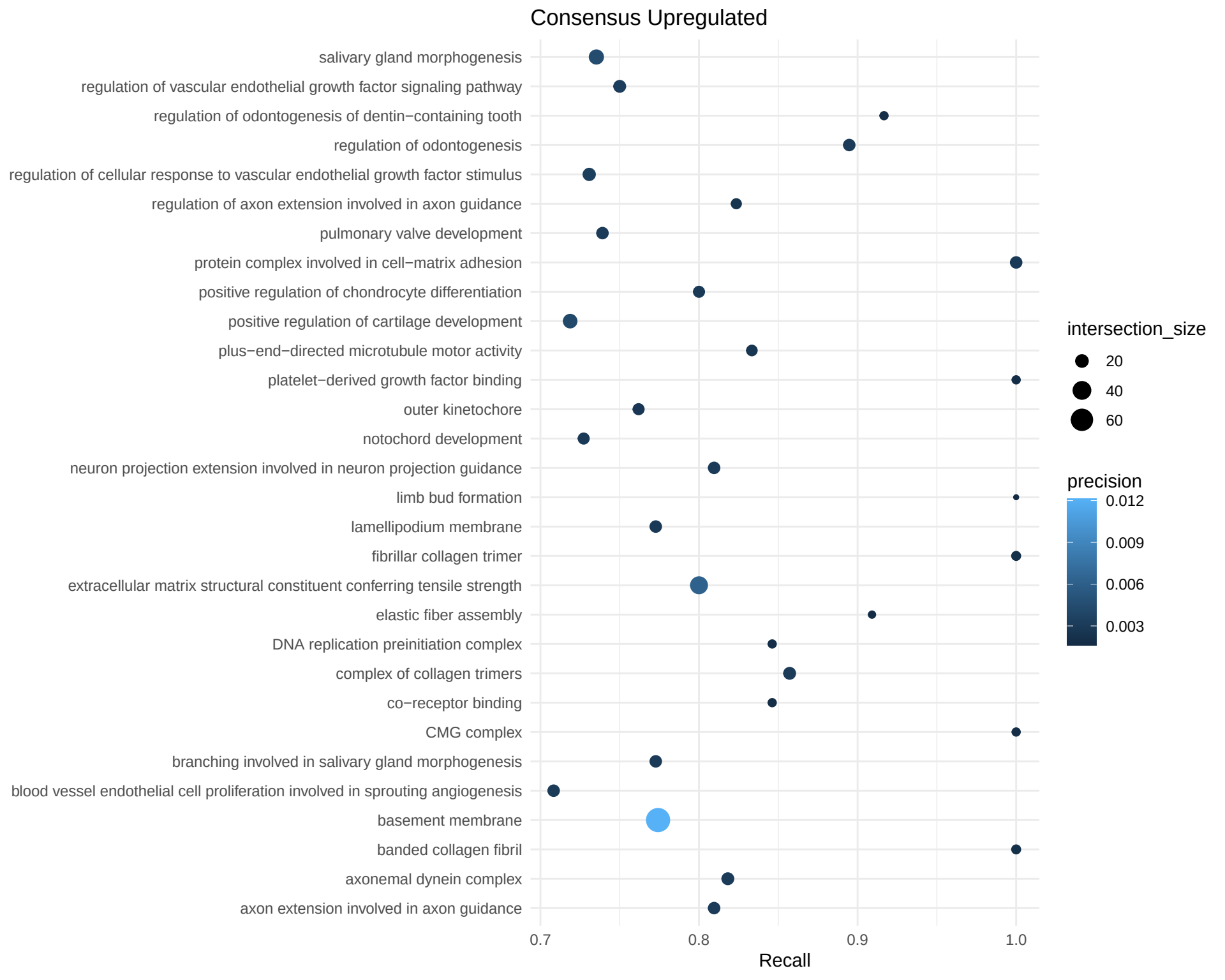
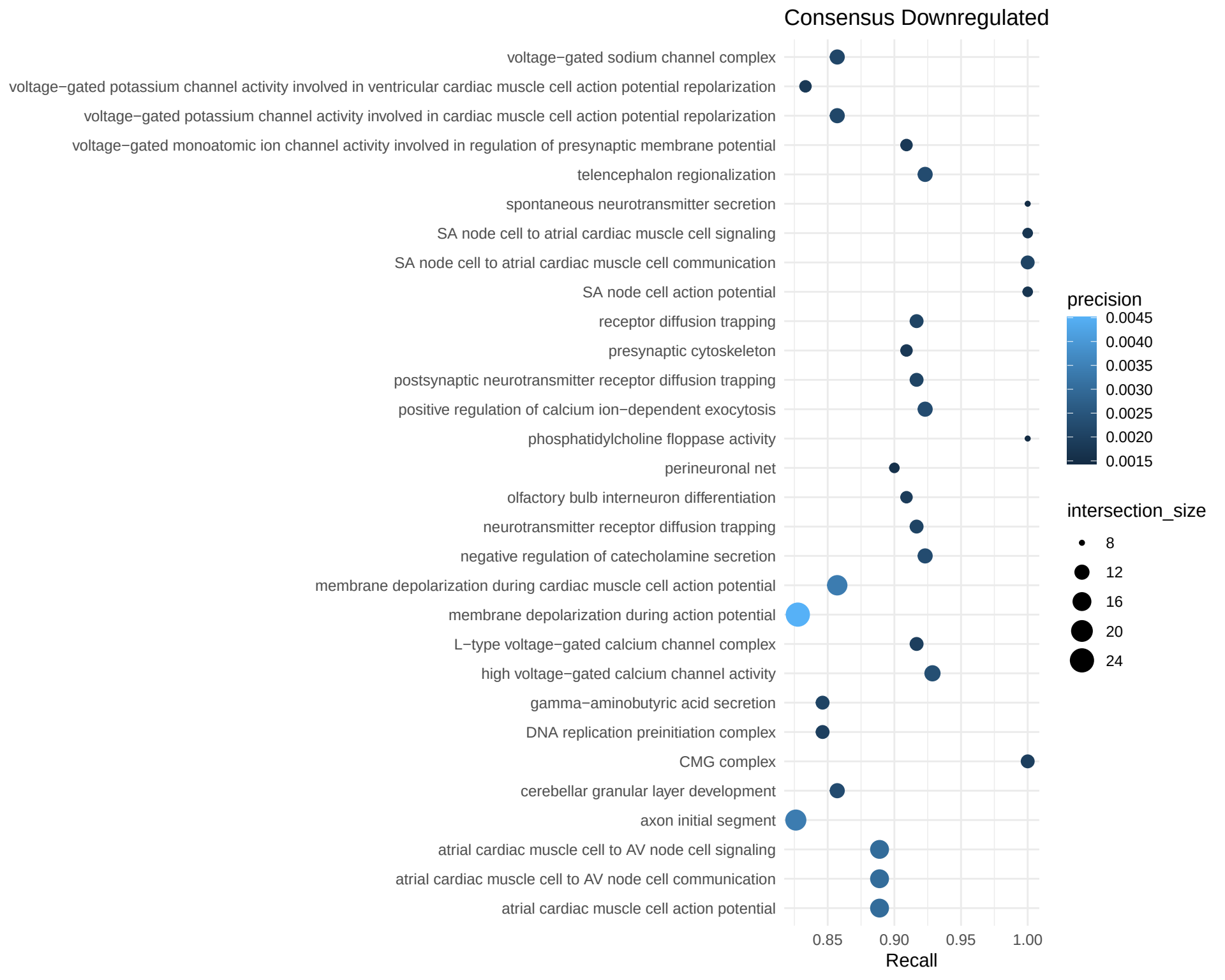
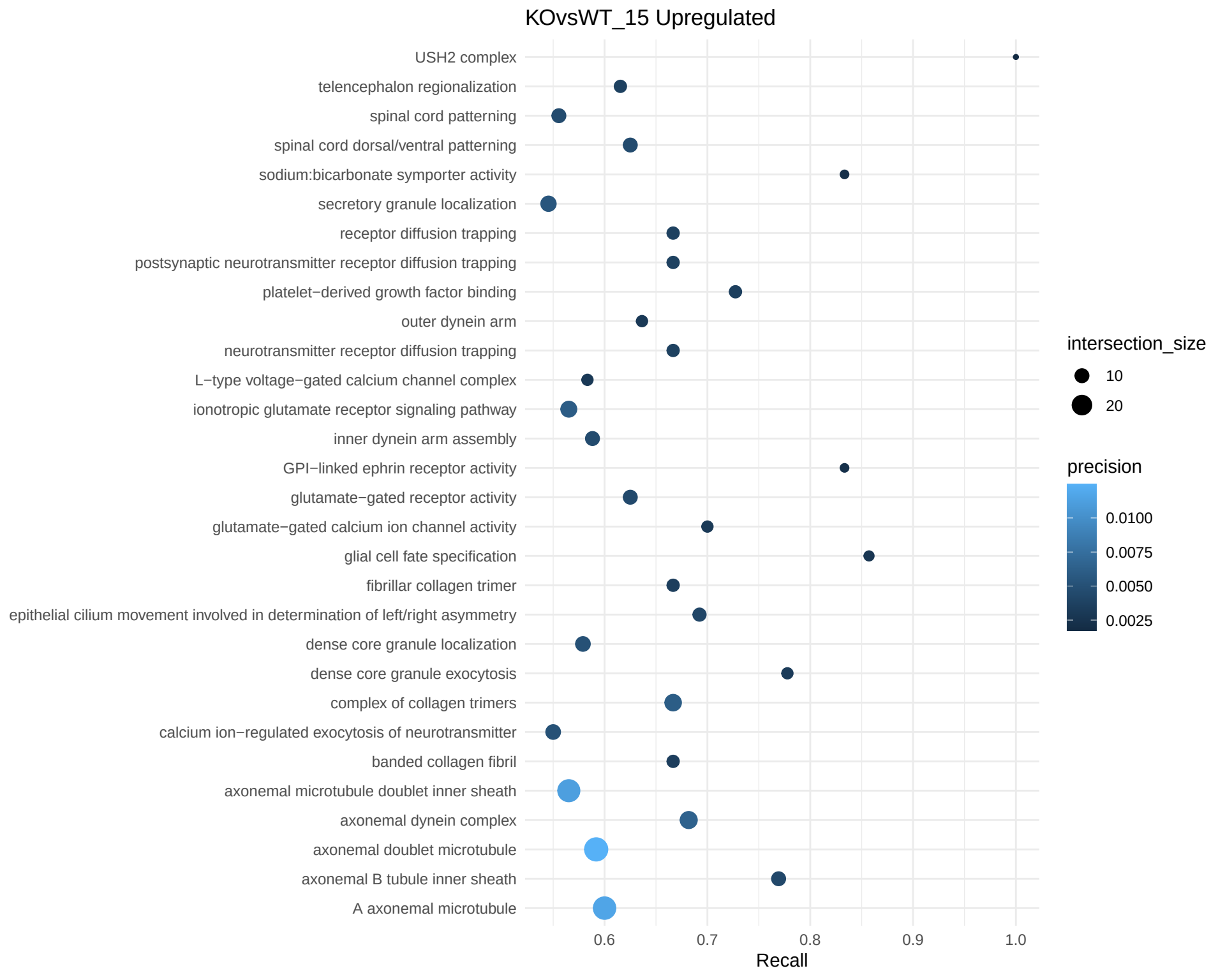
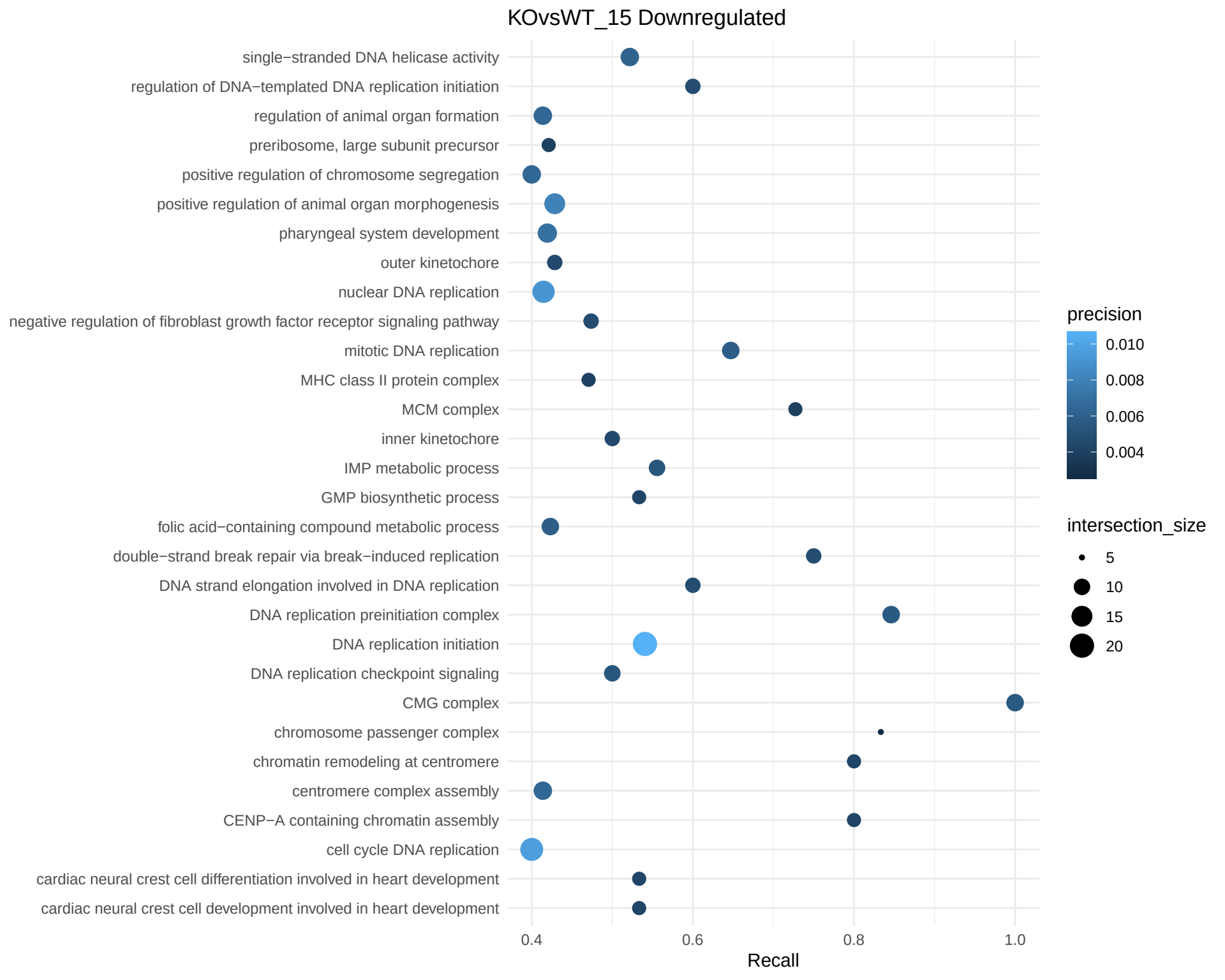
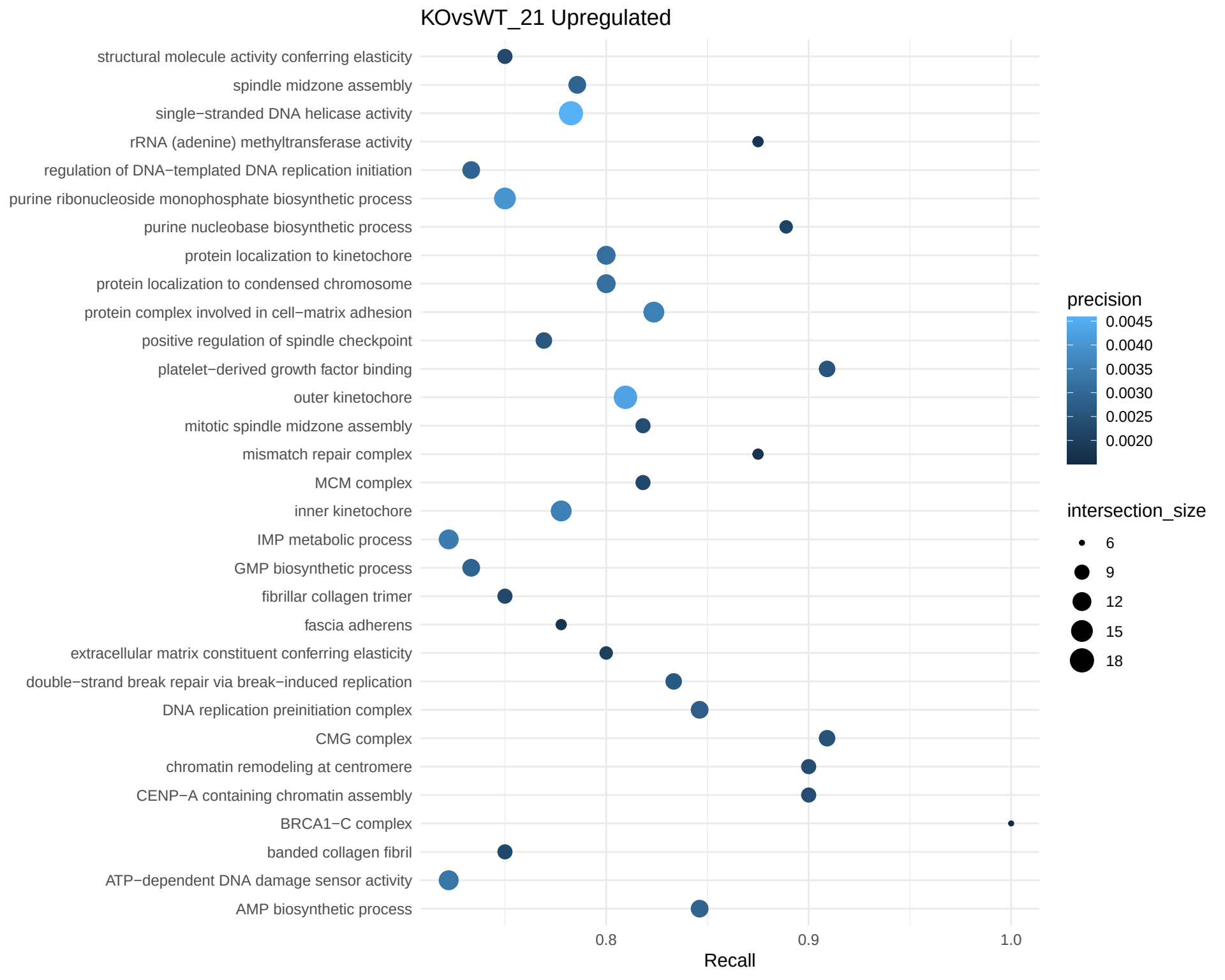
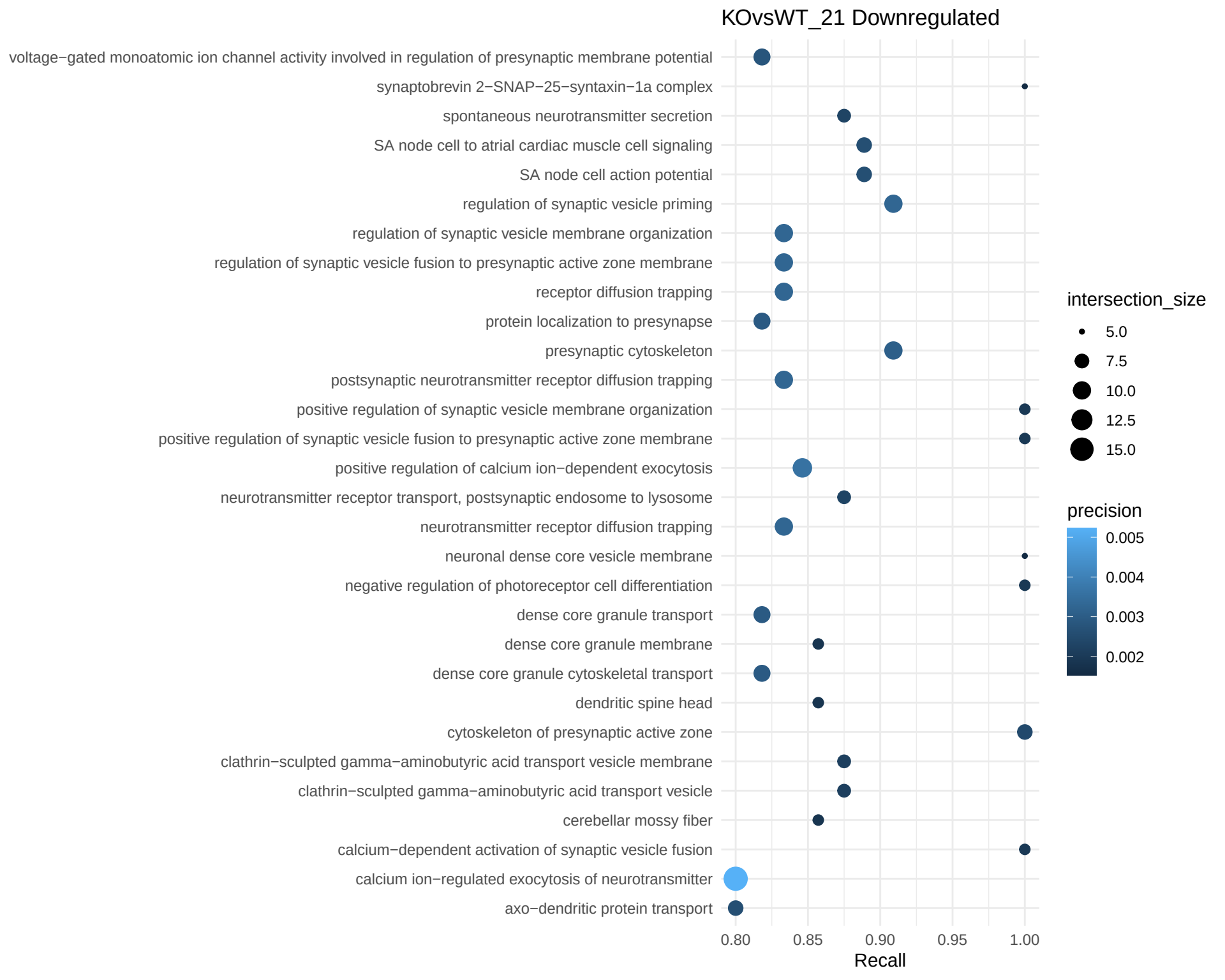

Consensus Upregulated
salivary gland morphogenesis
regulation of vascular endothelial growth factor signaling pathway
regulation of odontogenesis of dentin−containing tooth
regulation of odontogenesis
regulation of cellular response to vascular endothelial growth factor stimulus
regulation of axon extension involved in axon guidance
pulmonary valve development
protein complex involved in cell−matrix adhesion
positive regulation of chondrocyte differentiation
positive regulation of cartilage development
plus−end−directed microtubule motor activity
platelet−derived growth factor binding
outer kinetochore
notochord development
neuron projection extension involved in neuron projection guidance
limb bud formation
lamellipodium membrane
fibrillar collagen trimer
extracellular matrix structural constituent conferring tensile strength
elastic fiber assembly
DNA replication preinitiation complex
complex of collagen trimers
co−receptor binding
CMG complex
branching involved in salivary gland morphogenesis
blood vessel endothelial cell proliferation involved in sprouting angiogenesis
basement membrane
banded collagen fibril
axonemal dynein complex
axon extension involved in axon guidance
0.7
0.8
0.9
1.0
Recall
intersection_size
20
40
60
precision
0.012
0.009
0.006
0.003

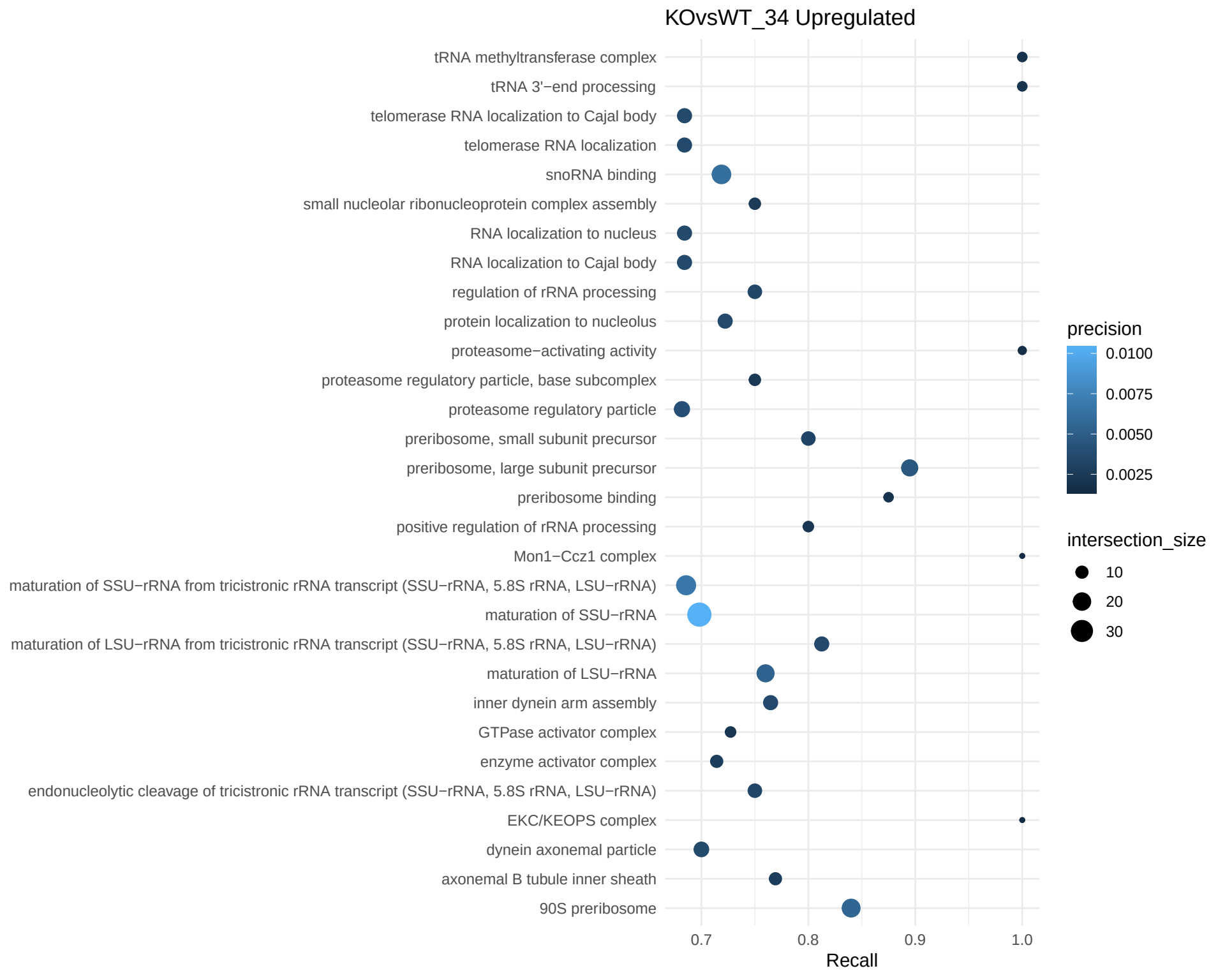

Consensus Downregulated
voltage−gated sodium channel complex
voltage−gated potassium channel activity involved in ventricular cardiac muscle cell action potential repolarization
voltage−gated potassium channel activity involved in cardiac muscle cell action potential repolarization
voltage−gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential
telencephalon regionalization
spontaneous neurotransmitter secretion
SA node cell to atrial cardiac muscle cell signaling
SA node cell to atrial cardiac muscle cell communication
SA node cell action potential
receptor diffusion trapping
presynaptic cytoskeleton
postsynaptic neurotransmitter receptor diffusion trapping
positive regulation of calcium ion−dependent exocytosis
phosphatidylcholine floppase activity
perineuronal net
olfactory bulb interneuron differentiation
neurotransmitter receptor diffusion trapping
negative regulation of catecholamine secretion
membrane depolarization during cardiac muscle cell action potential
membrane depolarization during action potential
L−type voltage−gated calcium channel complex
high voltage−gated calcium channel activity
gamma−aminobutyric acid secretion
DNA replication preinitiation complex
CMG complex
cerebellar granular layer development
axon initial segment
atrial cardiac muscle cell to AV node cell signaling
atrial cardiac muscle cell to AV node cell communication
atrial cardiac muscle cell action potential
0.85
0.90
0.95
1.00
Recall
precision
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
0.0015
intersection_size
8
12
16
20
24

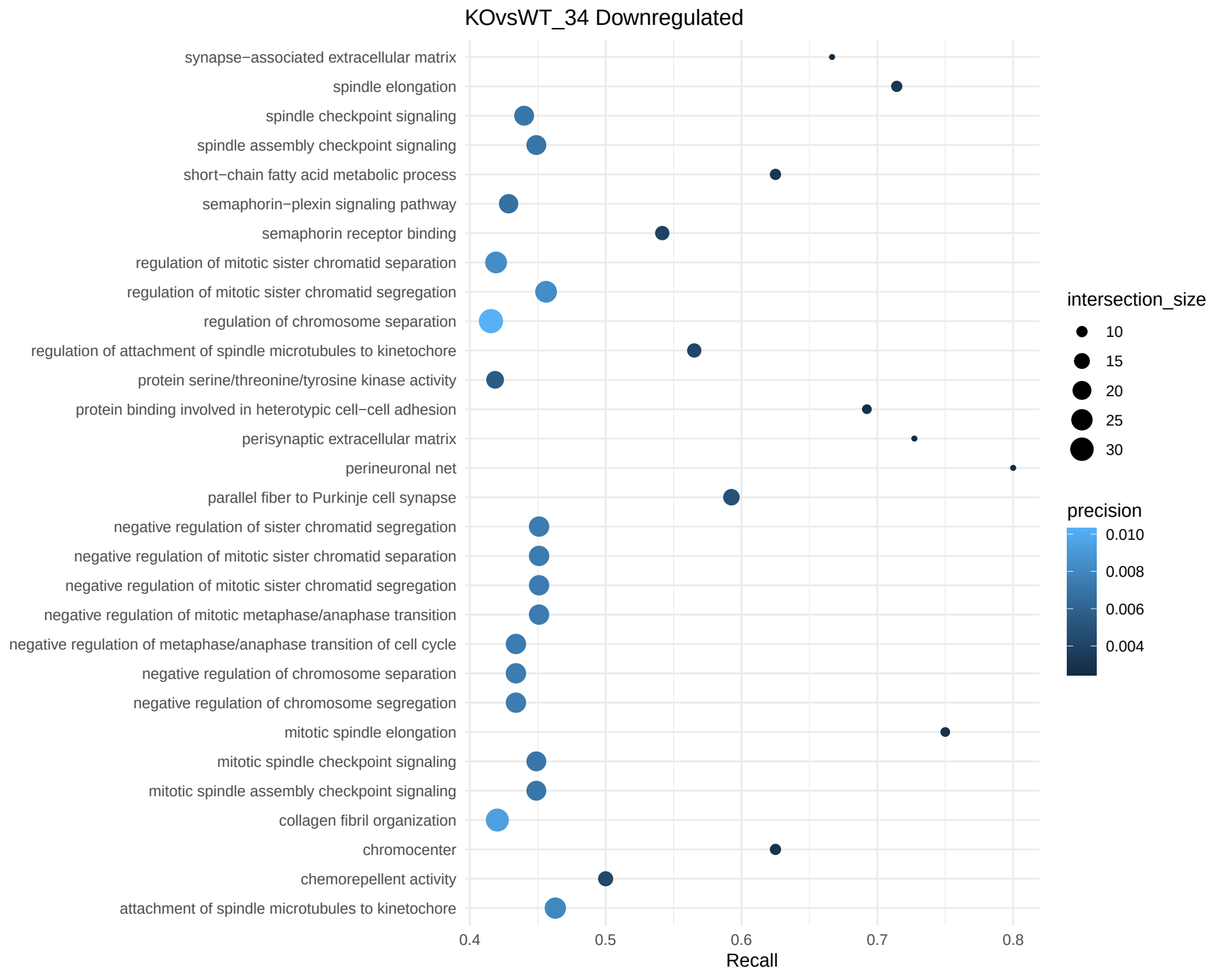

KOvsWT_15 Upregulated
USH2 complex
telencephalon regionalization
spinal cord patterning
spinal cord dorsal/ventral patterning
sodium:bicarbonate symporter activity
secretory granule localization
receptor diffusion trapping
postsynaptic neurotransmitter receptor diffusion trapping
platelet−derived growth factor binding
outer dynein arm
neurotransmitter receptor diffusion trapping
L−type voltage−gated calcium channel complex
ionotropic glutamate receptor signaling pathway
inner dynein arm assembly
GPI−linked ephrin receptor activity
glutamate−gated receptor activity
glutamate−gated calcium ion channel activity
glial cell fate specification
fibrillar collagen trimer
epithelial cilium movement involved in determination of left/right asymmetry
dense core granule localization
dense core granule exocytosis
complex of collagen trimers
calcium ion−regulated exocytosis of neurotransmitter
banded collagen fibril
axonemal microtubule doublet inner sheath
axonemal dynein complex
axonemal doublet microtubule
axonemal B tubule inner sheath
A axonemal microtubule
0.6
0.7
0.8
0.9
1.0
Recall
intersection_size
10
20
precision
0.0100
0.0075
0.0050
0.0025

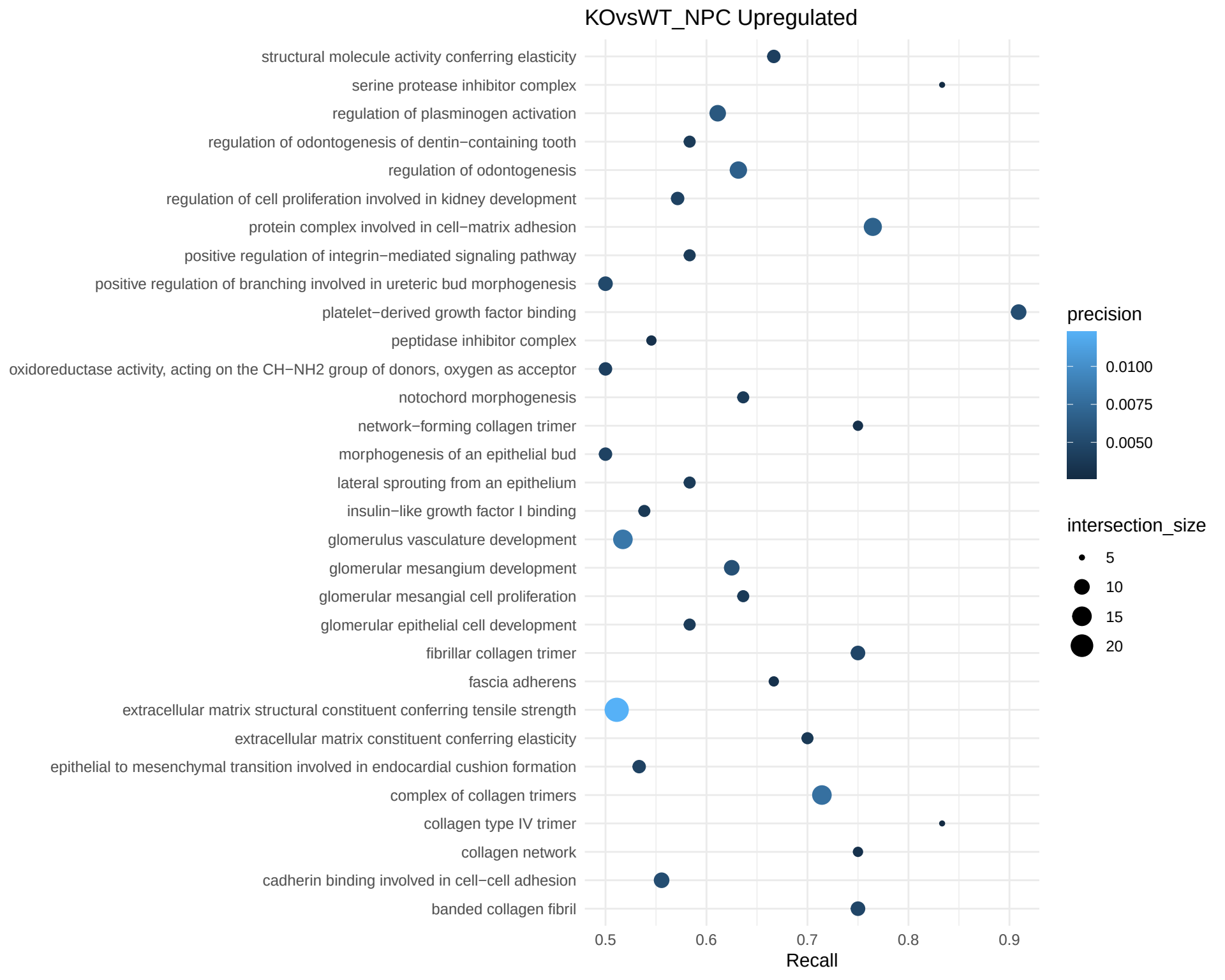

KOvsWT_15 Downregulated
single−stranded DNA helicase activity
regulation of DNA−templated DNA replication initiation
regulation of animal organ formation
preribosome, large subunit precursor
positive regulation of chromosome segregation
positive regulation of animal organ morphogenesis
pharyngeal system development
outer kinetochore
nuclear DNA replication
negative regulation of fibroblast growth factor receptor signaling pathway
mitotic DNA replication
MHC class II protein complex
MCM complex
inner kinetochore
IMP metabolic process
GMP biosynthetic process
folic acid−containing compound metabolic process
double−strand break repair via break−induced replication
DNA strand elongation involved in DNA replication
DNA replication preinitiation complex
DNA replication initiation
DNA replication checkpoint signaling
CMG complex
chromosome passenger complex
chromatin remodeling at centromere
centromere complex assembly
CENP−A containing chromatin assembly
cell cycle DNA replication
cardiac neural crest cell differentiation involved in heart development
cardiac neural crest cell development involved in heart development
0.4
0.6
0.8
1.0
Recall
precision
0.010
0.008
0.006
0.004
intersection_size
5
10
15
20

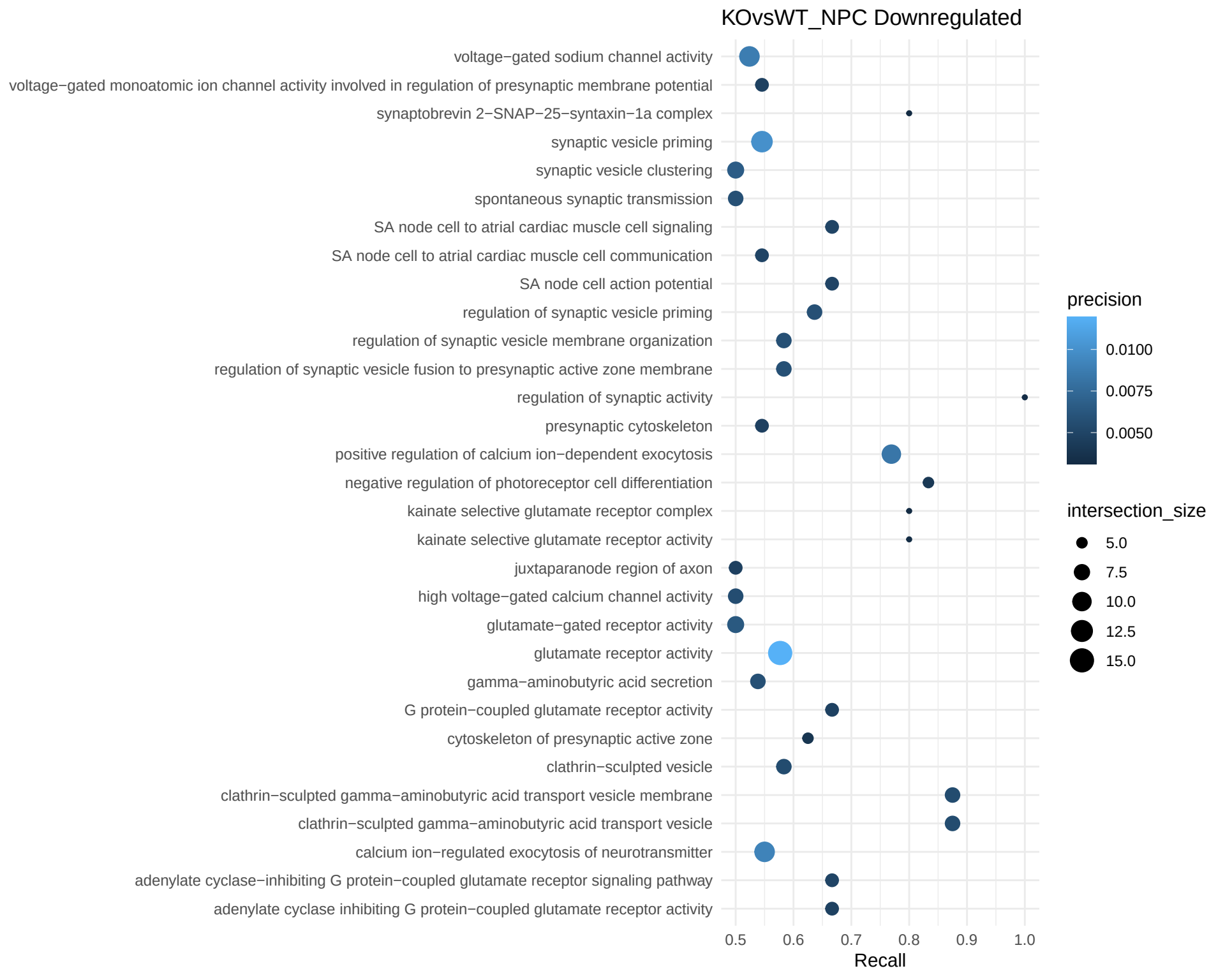

KOvsWT_21 Upregulated
structural molecule activity conferring elasticity
spindle midzone assembly
single−stranded DNA helicase activity
rRNA (adenine) methyltransferase activity
regulation of DNA−templated DNA replication initiation
purine ribonucleoside monophosphate biosynthetic process
purine nucleobase biosynthetic process
protein localization to kinetochore
protein localization to condensed chromosome
protein complex involved in cell−matrix adhesion
positive regulation of spindle checkpoint
platelet−derived growth factor binding
outer kinetochore
mitotic spindle midzone assembly
mismatch repair complex
MCM complex
inner kinetochore
IMP metabolic process
GMP biosynthetic process
fibrillar collagen trimer
fascia adherens
extracellular matrix constituent conferring elasticity
double−strand break repair via break−induced replication
DNA replication preinitiation complex
CMG complex
chromatin remodeling at centromere
CENP−A containing chromatin assembly
BRCA1−C complex
banded collagen fibril
ATP−dependent DNA damage sensor activity
AMP biosynthetic process
0.8
0.9
1.0
Recall
precision
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
intersection_size
6
9
12
15
18

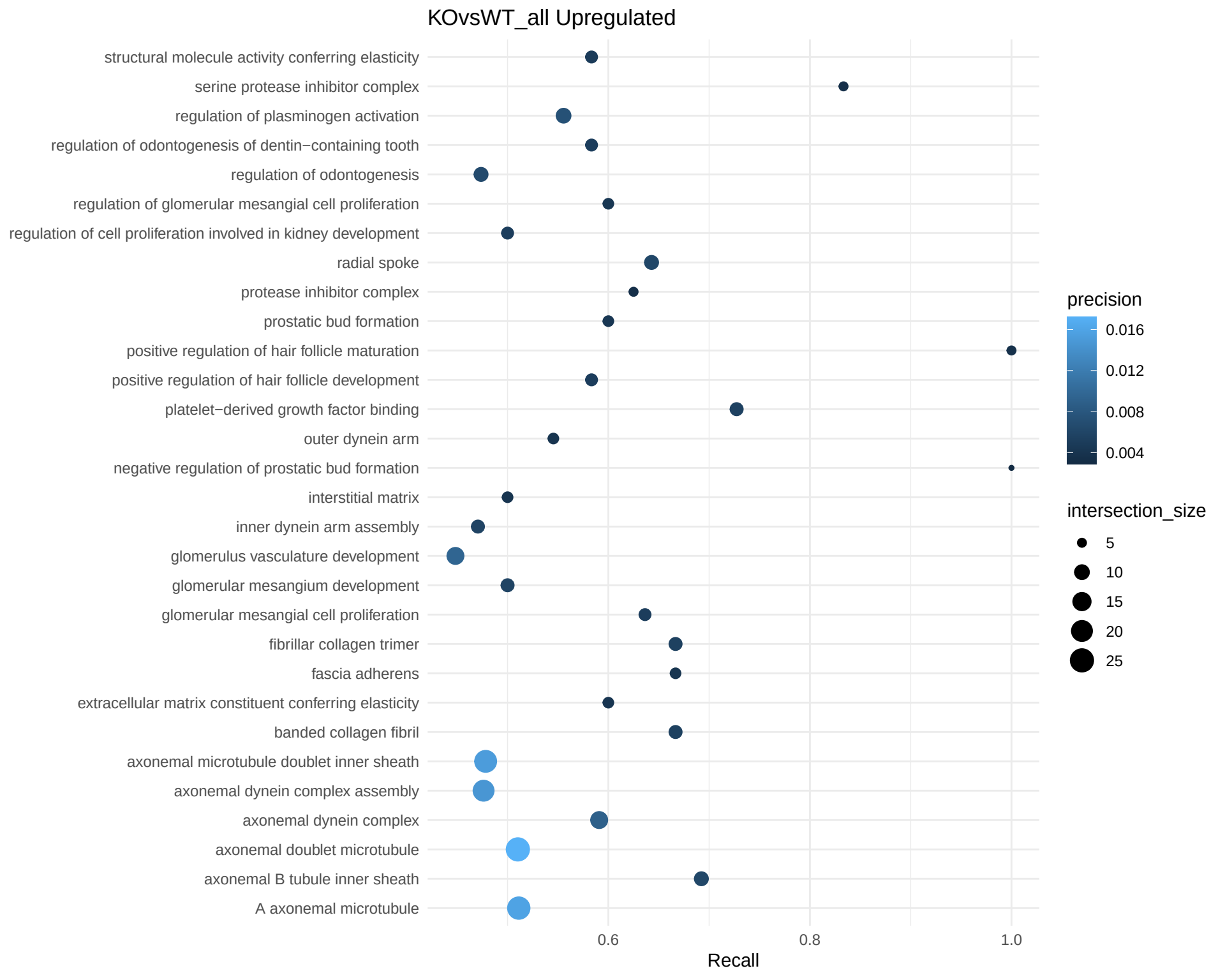

KOvsWT_21 Downregulated
voltage−gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential
synaptobrevin 2−SNAP−25−syntaxin−1a complex
spontaneous neurotransmitter secretion
SA node cell to atrial cardiac muscle cell signaling
SA node cell action potential
regulation of synaptic vesicle priming
regulation of synaptic vesicle membrane organization
regulation of synaptic vesicle fusion to presynaptic active zone membrane
receptor diffusion trapping
protein localization to presynapse
presynaptic cytoskeleton
postsynaptic neurotransmitter receptor diffusion trapping
positive regulation of synaptic vesicle membrane organization
positive regulation of synaptic vesicle fusion to presynaptic active zone membrane
positive regulation of calcium ion−dependent exocytosis
neurotransmitter receptor transport, postsynaptic endosome to lysosome
neurotransmitter receptor diffusion trapping
neuronal dense core vesicle membrane
negative regulation of photoreceptor cell differentiation
dense core granule transport
dense core granule membrane
dense core granule cytoskeletal transport
dendritic spine head
cytoskeleton of presynaptic active zone
clathrin−sculpted gamma−aminobutyric acid transport vesicle membrane
clathrin−sculpted gamma−aminobutyric acid transport vesicle
cerebellar mossy fiber
calcium−dependent activation of synaptic vesicle fusion
calcium ion−regulated exocytosis of neurotransmitter
axo−dendritic protein transport
0.80
0.85
0.90
0.95
1.00
Recall
intersection_size
5.0
7.5
10.0
12.5
15.0
precision
0.005
0.004
0.003
0.002

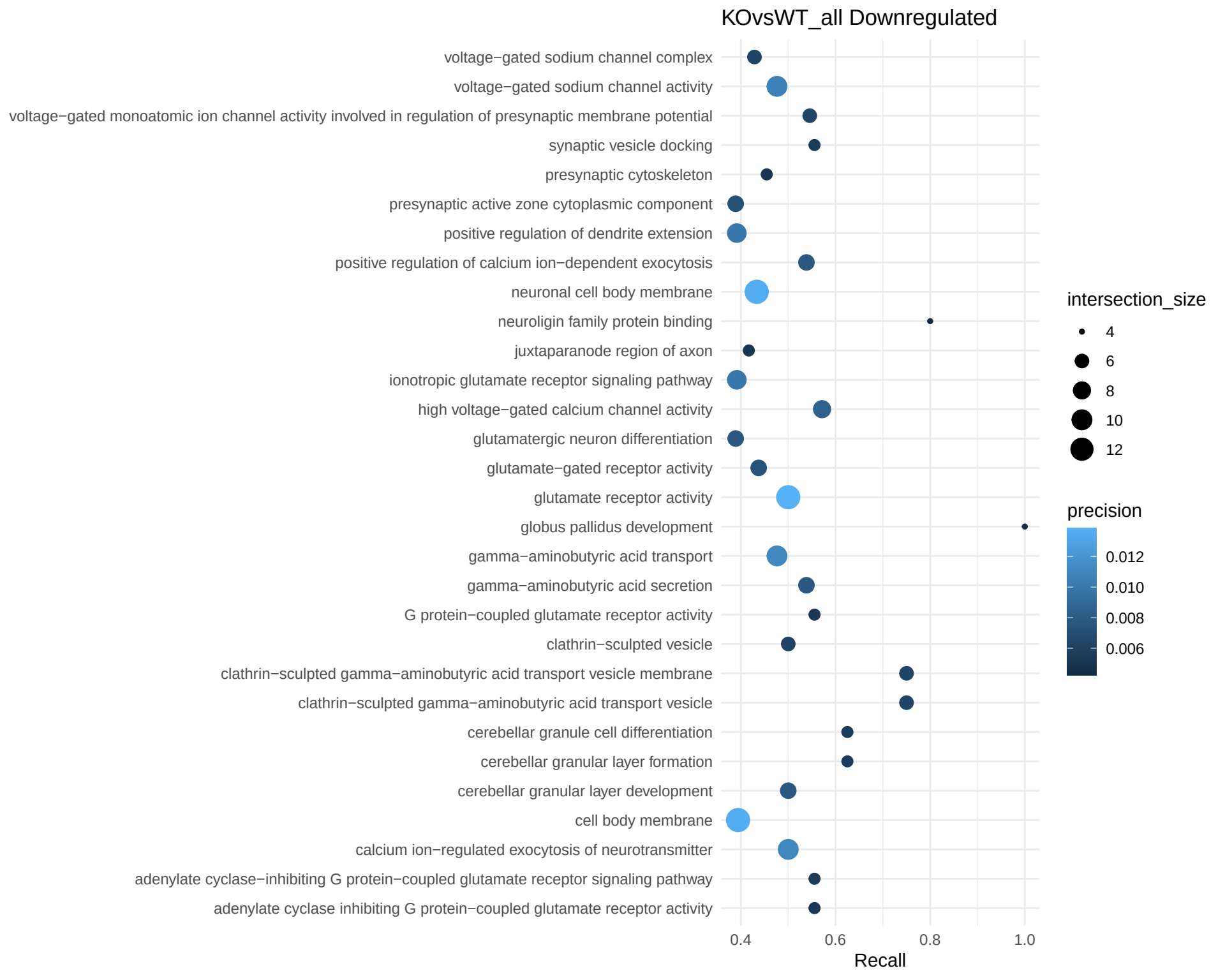

KOvsWT_34 Upregulated
tRNA methyltransferase complex
tRNA 3'−end processing
telomerase RNA localization to Cajal body
telomerase RNA localization
snoRNA binding
small nucleolar ribonucleoprotein complex assembly
RNA localization to nucleus
RNA localization to Cajal body
regulation of rRNA processing
protein localization to nucleolus
proteasome−activating activity
proteasome regulatory particle, base subcomplex
proteasome regulatory particle
preribosome, small subunit precursor
preribosome, large subunit precursor
preribosome binding
positive regulation of rRNA processing
Mon1−Ccz1 complex
maturation of SSU−rRNA from tricistronic rRNA transcript (SSU−rRNA, 5.8S rRNA, LSU−rRNA)
maturation of SSU−rRNA
maturation of LSU−rRNA from tricistronic rRNA transcript (SSU−rRNA, 5.8S rRNA, LSU−rRNA)
maturation of LSU−rRNA
inner dynein arm assembly
GTPase activator complex
enzyme activator complex
endonucleolytic cleavage of tricistronic rRNA transcript (SSU−rRNA, 5.8S rRNA, LSU−rRNA)
EKC/KEOPS complex
dynein axonemal particle
axonemal B tubule inner sheath
90S preribosome
0.7
0.8
0.9
1.0
Recall
precision
0.0100
0.0075
0.0050
0.0025
intersection_size
10
20
30


KOvsWT_34 Downregulated
synapse−associated extracellular matrix
spindle elongation
spindle checkpoint signaling
spindle assembly checkpoint signaling
short−chain fatty acid metabolic process
semaphorin−plexin signaling pathway
semaphorin receptor binding
regulation of mitotic sister chromatid separation
regulation of mitotic sister chromatid segregation
regulation of chromosome separation
regulation of attachment of spindle microtubules to kinetochore
protein serine/threonine/tyrosine kinase activity
protein binding involved in heterotypic cell−cell adhesion
perisynaptic extracellular matrix
perineuronal net
parallel fiber to Purkinje cell synapse
negative regulation of sister chromatid segregation
negative regulation of mitotic sister chromatid separation
negative regulation of mitotic sister chromatid segregation
negative regulation of mitotic metaphase/anaphase transition
negative regulation of metaphase/anaphase transition of cell cycle
negative regulation of chromosome separation
negative regulation of chromosome segregation
mitotic spindle elongation
mitotic spindle checkpoint signaling
mitotic spindle assembly checkpoint signaling
collagen fibril organization
chromocenter
chemorepellent activity
attachment of spindle microtubules to kinetochore
0.4
0.5
0.6
0.7
0.8
Recall
intersection_size
10
15
20
25
30
precision
0.010
0.008
0.006
0.004


KOvsWT_NPC Upregulated
structural molecule activity conferring elasticity
serine protease inhibitor complex
regulation of plasminogen activation
regulation of odontogenesis of dentin−containing tooth
regulation of odontogenesis
regulation of cell proliferation involved in kidney development
protein complex involved in cell−matrix adhesion
positive regulation of integrin−mediated signaling pathway
positive regulation of branching involved in ureteric bud morphogenesis
platelet−derived growth factor binding
peptidase inhibitor complex
oxidoreductase activity, acting on the CH−NH2 group of donors, oxygen as acceptor
notochord morphogenesis
network−forming collagen trimer
morphogenesis of an epithelial bud
lateral sprouting from an epithelium
insulin−like growth factor I binding
glomerulus vasculature development
glomerular mesangium development
glomerular mesangial cell proliferation
glomerular epithelial cell development
fibrillar collagen trimer
fascia adherens
extracellular matrix structural constituent conferring tensile strength
extracellular matrix constituent conferring elasticity
epithelial to mesenchymal transition involved in endocardial cushion formation
complex of collagen trimers
collagen type IV trimer
collagen network
cadherin binding involved in cell−cell adhesion
banded collagen fibril
0.5
0.6
0.7
0.8
0.9
Recall
precision
0.0100
0.0075
0.0050
intersection_size
5
10
15
20


KOvsWT_NPC Downregulated
voltage−gated sodium channel activity
voltage−gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential
synaptobrevin 2−SNAP−25−syntaxin−1a complex
synaptic vesicle priming
synaptic vesicle clustering
spontaneous synaptic transmission
SA node cell to atrial cardiac muscle cell signaling
SA node cell to atrial cardiac muscle cell communication
SA node cell action potential
regulation of synaptic vesicle priming
regulation of synaptic vesicle membrane organization
regulation of synaptic vesicle fusion to presynaptic active zone membrane
regulation of synaptic activity
presynaptic cytoskeleton
positive regulation of calcium ion−dependent exocytosis
negative regulation of photoreceptor cell differentiation
kainate selective glutamate receptor complex
kainate selective glutamate receptor activity
juxtaparanode region of axon
high voltage−gated calcium channel activity
glutamate−gated receptor activity
glutamate receptor activity
gamma−aminobutyric acid secretion
G protein−coupled glutamate receptor activity
cytoskeleton of presynaptic active zone
clathrin−sculpted vesicle
clathrin−sculpted gamma−aminobutyric acid transport vesicle membrane
clathrin−sculpted gamma−aminobutyric acid transport vesicle
calcium ion−regulated exocytosis of neurotransmitter
adenylate cyclase−inhibiting G protein−coupled glutamate receptor signaling pathway
adenylate cyclase inhibiting G protein−coupled glutamate receptor activity
0.5
0.6
0.7
0.8
0.9
1.0
Recall
precision
0.0100
0.0075
0.0050
intersection_size
5.0
7.5
10.0
12.5
15.0


KOvsWT_all Upregulated
structural molecule activity conferring elasticity
serine protease inhibitor complex
regulation of plasminogen activation
regulation of odontogenesis of dentin−containing tooth
regulation of odontogenesis
regulation of glomerular mesangial cell proliferation
regulation of cell proliferation involved in kidney development
radial spoke
protease inhibitor complex
prostatic bud formation
positive regulation of hair follicle maturation
positive regulation of hair follicle development
platelet−derived growth factor binding
outer dynein arm
negative regulation of prostatic bud formation
interstitial matrix
inner dynein arm assembly
glomerulus vasculature development
glomerular mesangium development
glomerular mesangial cell proliferation
fibrillar collagen trimer
fascia adherens
extracellular matrix constituent conferring elasticity
banded collagen fibril
axonemal microtubule doublet inner sheath
axonemal dynein complex assembly
axonemal dynein complex
axonemal doublet microtubule
axonemal B tubule inner sheath
A axonemal microtubule
0.6
0.8
1.0
Recall
precision
0.016
0.012
0.008
0.004
intersection_size
5
10
15
20
25


KOvsWT_all Downregulated
voltage−gated sodium channel complex
voltage−gated sodium channel activity
voltage−gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential
synaptic vesicle docking
presynaptic cytoskeleton
presynaptic active zone cytoplasmic component
positive regulation of dendrite extension
positive regulation of calcium ion−dependent exocytosis
neuronal cell body membrane
neuroligin family protein binding
juxtaparanode region of axon
ionotropic glutamate receptor signaling pathway
high voltage−gated calcium channel activity
glutamatergic neuron differentiation
glutamate−gated receptor activity
glutamate receptor activity
globus pallidus development
gamma−aminobutyric acid transport
gamma−aminobutyric acid secretion
G protein−coupled glutamate receptor activity
clathrin−sculpted vesicle
clathrin−sculpted gamma−aminobutyric acid transport vesicle membrane
clathrin−sculpted gamma−aminobutyric acid transport vesicle
cerebellar granule cell differentiation
cerebellar granular layer formation
cerebellar granular layer development
cell body membrane
calcium ion−regulated exocytosis of neurotransmitter
adenylate cyclase−inhibiting G protein−coupled glutamate receptor signaling pathway
adenylate cyclase inhibiting G protein−coupled glutamate receptor activity
0.4
0.6
0.8
1.0
Recall
intersection_size
4
6
8
10
12
precision
0.012
0.010
0.008
0.006